SOAP: Credit Risk Model for Lending Decision

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Executive Decision and Performance Transformation

The rebuilt credit risk model substantially improves applicant ranking compared to the legacy benchmark. The legacy model demonstrated poor discriminatory power (validation Gini of **0.110**, AUC of **0.555**), which was inadequate for retail banking operations without significant redevelopment. On common validation data, the rebuilt Generalized Linear Model (GLM) increased the Gini coefficient by **+0.286 points** to **0.396**, while tests on untouched historical applications confirmed stable performance with a Gini of **0.374** and AUC of **0.687**.

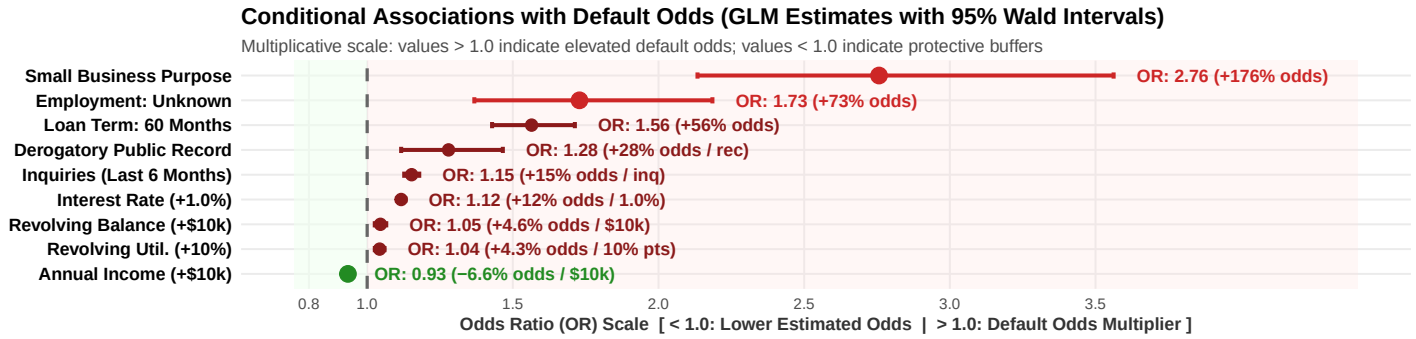
Recommendation: Adopt the transparent GLM as the candidate applicant scoring engine, subject to recalibration on contemporary Australian banking data and independent validation. Reserve the spatial-temporal Generalized Linear Mixed Model (GLMM) for portfolio risk monitoring rather than live customer underwriting.

Methodology: Binomial logistic regression, stepwise selection via Akaike Information Criterion, stratified 5-fold cross-validation, held-out temporal validation, and mixed-effects modelling incorporating state-level and origination-month random effects.

Performance Metric	Legacy Benchmark	Rebuilt Model	Operational Takeaway
Validation Discrimination (Gini)	0.110 (AUC = 0.555)	0.396 (AUC = 0.698)	+0.286 validation Gini points on common evaluation sample
Held-Out Test Sample	Not available	Gini 0.374; AUC 0.687	95% CI: AUC [0.670, 0.703]; performance remained stable on untouched test sample
Maximum Separation (KS)	Not available	KS = 0.280 (at 40.1% volume)	Maximum separation between cumulative default and non-default scores at 40.1% volume
Illustrative Policy Scenario (15% Cutoff)	Not applicable	60.6% approval volume	Approved default rate: 9.2% (vs 14.9% baseline); captures 62.5% of defaulting accounts

Primary Risk Drivers: Conditional Associations with Default Odds

To support regulatory explainability and adverse-action notices, the scorecard estimates conditional Odds Ratios with 95% Wald confidence intervals. Values above 1.0 indicate elevated default odds, whereas values below 1.0 represent protective financial buffers.

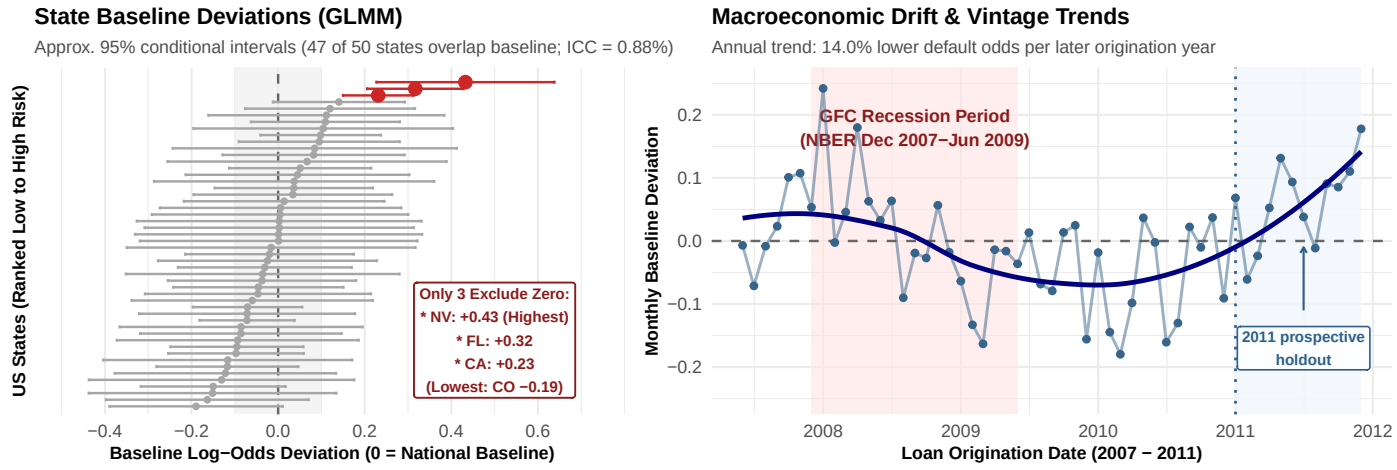


Loan purpose and credit tenure emerge as the primary risk discriminators. Small-business loans carry the highest risk, with default odds 2.76 times higher than car loans (Odds Ratio 2.76, 95% confidence interval [2.13, 3.56]). Unreported employment tenure and 60-month loan terms also significantly increase risk (elevating odds by 73% and 56%, respectively). Additional financial indicators, including higher interest rates, credit inquiries, derogatory public records, revolving balances, and credit line utilization, all consistently raise default probability. Conversely, higher annual income provides the primary buffer, reducing default odds by 6.6% per additional $10,000 earned.

Traditional measures such as debt-to-income and historical delinquency were not selected because income, revolving debt, utilization, and recent inquiries capture repayment ability with greater precision. While excluded from the statistical scoring formula, prudential guidelines (APRA APS 220) still require maintaining policy limits on debt-to-income and delinquency for all applicants.

Geographic and Vintage Dynamics: The "Ranking versus Monitoring" Paradox

Incorporating geographic regions and origination timing helps portfolio management track macroeconomic cycles, but does not improve individual applicant ranking:



When evaluated on forward holdout applications from 2011, incorporating state and vintage effects produced no meaningful gain in applicant ranking: the standard GLM achieved a Gini of 0.4070 compared to 0.4069 for the mixed-effects model. Individual borrower characteristics accounted for essentially all discriminatory power. Crucially, a side-by-side comparison revealed no surprising differences or reversals in variable importance once geographic and temporal factors were controlled: odds ratios for all continuous risk drivers shifted by less than 1.5%. This parameter stability confirms that individual creditworthiness operates consistently across jurisdictions and economic vintages.

Time was explored through both a continuous annual trend and monthly cohort adjustments. Holding borrower characteristics constant, later origination years showed 14% lower default odds annually (Odds Ratio 0.860, 95% CI [0.798, 0.927]), reflecting evolving underwriting standards and economic conditions. Across jurisdictions, state differences represented less than 1% of total risk variation, with only Nevada, Florida, and California showing baseline default rates that statistically exceeded the national average. However, standard errors for regional estimates tend to understate true uncertainty because they treat overall portfolio variance as fixed. Consequently, management must not introduce regional lending quotas or geographic risk pricing, avoiding substantial fairness, legal, and regulatory concerns.

Diagnostic Findings, Governance Limitations, and Deployment Roadmap

Model diagnostic checks confirmed that overall residual distributions and dispersion align with theoretical standards, with low collinearity among predictors (adjusted GVIF below 1.6). In large datasets exceeding 30,000 loans, formal goodness-of-fit significance tests become overly sensitive to trivial deviations that lack practical consequence. Evaluating loans across risk deciles demonstrated reliable rank ordering, though mild underprediction occurred among intermediate risk bands. Additionally, monthly aggregated residuals exhibited moderate serial persistence between adjacent origination cohorts, reflecting cyclical macroeconomic momentum rather than omitted borrower characteristics.

A central operational hurdle for time-dependent models in production is the future vintage dilemma: incoming loan applicants in future periods cannot have historical adjustments assigned to them in advance. Assuming an incoming cohort represents an average economic state fails during unexpected downturns and requires mathematical adjustments to avoid probability distortion, whereas statistical forecasting injects out-of-sample volatility into credit decisions. In production, risk management should replace latent time effects with observable macroeconomic covariates, such as official cash rates, inflation, and unemployment figures. This operational boundary reinforces our recommended dual-model architecture: the transparent applicant model delivers deterministic, explainable daily underwriting, while the mixed-effects framework is retained for retrospective portfolio monitoring and quarterly stress testing.

Before deployment in an Australian retail banking environment, the scorecard requires retrained calibration against domestic credit data, incorporating Comprehensive Credit Reporting (CCR) histories and Household Expenditure Measure (HEM) affordability benchmarks. Model governance under APRA APS 220 requires establishing ongoing tracking metrics, including the Population Stability Index and Characteristic Stability Index, to detect population drift and trigger timely recalibration. Furthermore, because this scorecard estimates Probability of Default (PD), regulatory capital provisioning under Basel frameworks necessitates integrating validated Loss Given Default (LGD) and Exposure at Default (EAD) models. Finally, the linear scoring structure supports compliant adverse-action reporting, ensuring clear communication of the principal reasons for any declined applications.